Master Thesis AI Review

Document: molecular-transformer-a-model-for-uncertainty-calibrated-chemical-reaction-prediction.pdf

Total Sections Reviewed: 9

Overall Scores:
Clarity: 8.1/10 | Logic: 9.0/10 | Rigor: 8.0/10

Section 1: Chemical Reaction Prediction

Metric	Score	Visual
Clarity	9/10	■■■■■■■■■■
Logic	9/10	■■■■■■■■■■
Rigor	7/10	■■■■■■■■■■

Summary:

This is a strong, well-structured abstract that effectively communicates the research problem, the proposed method, and its significant results. Its primary weaknesses are a few instances of overstatement that could be tempered to enhance scientific credibility.

Critical Issues:

• The abstract makes several absolute claims ('outperforms all algorithms', 'universally applicable') that are difficult to substantiate. These should be moderated to reflect a more cautious and precise academic tone (e.g., 'outperforms current state-of-the-art methods', 'broadly applicable').
• The description of the uncertainty metric's accuracy is slightly convoluted and could be rephrased for greater clarity and impact.

Suggested Edits:

Edit #1 | **Tone**

Original: Organic synthesis is one of the key stumbling blocks in medicinal chemistry.

Suggested: Planning organic synthesis remains a primary challenge in medicinal chemistry.

Reason: While 'stumbling blocks' is understandable, 'primary challenge' is more formal and standard in academic writing, setting a more professional tone from the outset.

Edit #2 | **Clarity**

Original: A necessary yet unsolved step in planning synthesis is solving the forward problem: Given reactants and reagents, predict the products.

Suggested: A critical and challenging step in synthesis planning is the forward problem: predicting the products given a set of reactants and reagents.

Reason: The term 'unsolved' can be too absolute. 'Challenging' is more accurate as partial solutions exist. The rephrasing improves flow and conciseness.

Edit #3 | **Grammar**

Original: Similar to other work, we treat reaction prediction as a machine translation problem between simpli ■ ed molecular-input line-entry system (SMILES) strings...

Suggested: Building on previous work, we frame reaction prediction as a machine translation problem between simplified molecular-input line-entry system (SMILES) strings...

Reason: 'Building on previous work' is a more conventional academic phrase. This revision also corrects a typographical error ('simpli ■ ed') where a space was inserted.

Edit #4 | Tone

Original: We show that a multihead attention Molecular Transformer model outperforms all algorithms in the literature, achieving a top-1 accuracy above 90% on a common benchmark data set.

Suggested: We demonstrate that a multihead attention Molecular Transformer model outperforms previously reported algorithms, achieving a top-1 accuracy exceeding 90% on a common benchmark dataset.

Reason: The claim to outperform 'all algorithms in the literature' is an overstatement. 'Previously reported algorithms' is more precise and academically defensible. 'Demonstrate' is a strong alternative to 'show', and 'exceeding' is more formal than 'above'.

Edit #5 | Clarity

Original: Molecular Transformer makes predictions by inferring the correlations between the presence and absence of chemical motifs in the reactant, reagent, and product present in the data set.

Suggested: The Molecular Transformer makes predictions by inferring correlations between chemical motifs present in the reactants, reagents, and products within the training dataset.

Reason: The original phrasing 'product present in the data set' is slightly awkward. The revision clarifies that the model learns from correlations across all components within the dataset.

Edit #6 | Clarity

Original: Crucially, our model can accurately estimate its own uncertainty, with an uncertainty score that is 89% accurate in terms of classifying whether a prediction is correct.

Suggested: Crucially, the model accurately estimates its own uncertainty; its confidence score correctly classifies a prediction's validity with 89% accuracy.

Reason: This revision simplifies the convoluted phrasing. It more directly states that the model's confidence in a prediction is a reliable indicator of its correctness.

Edit #7 | Tone

Original: ...which makes our method universally applicable.

Suggested: ...which significantly broadens the applicability of our method.

Reason: 'Universally applicable' is a hyperbolic claim that is nearly impossible to defend. 'Significantly broadens the applicability' is a strong but more realistic and credible assertion for academic research.

Section 2: ■ INTRODUCTION

Metric	Score	Visual
Clarity	9/10	■■■■■■■■■■
Logic	10/10	■■■■■■■■■■
Rigor	9/10	■■■■■■■■■■

Summary:

This is an exceptionally well-structured and compelling introduction that effectively establishes the research problem, critically reviews the state-of-the-art, and clearly positions the author's contribution. Its primary strength lies in the logical 'funnel' structure that guides the reader from a broad challenge to a specific, well-motivated research gap. Minor revisions should focus on refining certain phrases to enhance the formal academic tone and ensure maximum precision.

Critical Issues:

• The logical structure is already excellent and requires no critical changes. The main focus for revision should be on elevating the academic tone by replacing colloquialisms (e.g., 'stumbling blocks', 'vicious circle') with more formal terminology.
• The use of the word 'significant' should be reserved for statistical significance. In its current context, consider replacing it with 'substantial' or 'considerable' to avoid ambiguity.

Suggested Edits:

Edit #1 | **Tone**

Original: remains one of the key stumbling blocks in drug discovery.

Suggested: remains one of the key challenges in drug discovery.

Reason: The term 'stumbling blocks' is slightly colloquial. 'Challenges' or 'impediments' is more formal academic language.

Edit #2 | **Tone**

Original: generative models in machine learning for molecules are coming of age.

Suggested: generative models in machine learning for molecules are reaching maturity.

Reason: 'Coming of age' is an idiomatic expression. 'Reaching maturity' or 'becoming increasingly sophisticated' maintains the meaning in a more academic tone.

Edit #3 | **Tone**

Original: strategies that involve handcrafted rules quickly become intractable.

Suggested: strategies that rely on manually curated rules quickly become intractable.

Reason: 'Handcrafted' can imply artistry. 'Manually curated' or 'expert-defined' are more precise terms in a scientific context.

Edit #4 | **Grammar**

Original: dependent on the what other functional groups are nearby.

Suggested: dependent on which other functional groups are nearby.

Reason: Corrects a minor grammatical error ('the what' to 'which').

Edit #5 | **Tone**

Original: put the riskier steps in the beginning of the synthesis so that one can fail fast and fail cheap.

Suggested: prioritize riskier steps early in the synthesis to enable rapid and cost-effective invalidation of unpromising routes.

Reason: Replaces the colloquial business mantra 'fail fast and fail cheap' with a more formal and descriptive explanation of the scientific principle.

Edit #6 | **Clarity**

Original: The focus of the literature has thus far been on the latter question of predicting whether a template applies.

Suggested: Research has thus far focused primarily on predicting whether a given template applies.

Reason: This revision uses a more active construction ('Research has focused') and is more direct and concise.

Edit #7 | **Tone**

Original: This creates a vicious circle. Atom-mapping is based on templates and templates are based on atom mapping...

Suggested: This creates a circular dependency. Atom-mapping algorithms rely on templates, while template extraction relies on pre-existing atom maps...

Reason: 'Vicious circle' is an idiom. 'Circular dependency' or 'dependency loop' is a more precise and formal term for this logical problem.

Edit #8 | **Tone**

Original: seemingly automatic techniques are actually premised on handcrafted and often artisanal chemical rules.

Suggested: seemingly automatic techniques are ultimately premised on manually curated and often heuristic chemical rules.

Reason: Replaces 'handcrafted' and the highly informal 'artisanal' with the more standard 'manually curated' and 'heuristic'.

Edit #9 | **Tone**

Original: the accuracy was significantly improved by excluding the reagents from the reactivity score prediction

Suggested: the accuracy was substantially improved by excluding the reagents from the reactivity score prediction

Reason: In academic writing, 'significant' implies statistical significance. 'Substantially' or 'considerably' should be used to denote a large but non-statistical improvement.

Edit #10 | **Tone**

Original: A side effect of phrasing reaction prediction as predicting electron flow is that...

Suggested: A consequence of formulating reaction prediction as the prediction of electron flow is that...

Reason: 'Formulating' is a more precise and academic verb than 'phrasing', and 'consequence' is slightly more formal than 'side effect'.

Edit #11 | Tone

Original: getting the correct chemoselectivity, regioselectivity, and, to some extent, stereoselectivity.

Suggested: achieving the correct chemoselectivity, regioselectivity, and, to some extent, stereoselectivity.

Reason: The verb 'achieving' is more formal and appropriate than 'getting' in this scientific context.

Edit #12 | Clarity

Original: Our model has been made available since August 2018 in the backend of the IBM RXN for Chemistry, a free web-based graphical user interface, and has been used by several thousand organic chemists worldwide to perform more than 40 000 predictions so far.

Suggested: Since August 2018, the model has been implemented as the backend for the IBM RXN for Chemistry platform. This publicly available tool has been used by chemists worldwide to perform over 40,000 predictions.

Reason: This revision adopts a more active voice ('the model has been implemented') and improves the flow by breaking the long sentence into two, making the impressive usage statistics stand out more clearly.

Section 3: ■ DATA

Metric	Score	Visual
Clarity	8/10	■■■■■■■■■■
Logic	9/10	■■■■■■■■■■
Rigor	9/10	■■■■■■■■■■

Summary:

This section provides a well-structured and scientifically rigorous overview of the datasets and preprocessing methods. It effectively justifies its methodological choices by referencing prior work and explaining the rationale for its novel 'mixed' preprocessing approach. While the content is strong, the prose could be tightened for greater conciseness and a more formal academic tone.

Critical Issues:

• The narrative flow could be improved by framing the choice of the 'mixed' preprocessing method more proactively as a solution to an existing problem, rather than a decision made due to an unfortunate ambiguity.
• While logical, the first paragraph introduces four datasets and their interrelations in a very dense manner. Consider splitting this into two paragraphs: one for the patent-derived datasets (USPTO_MIT, LEF, STEREO) and another for the nonpublic Pistachio set, which would improve readability.

Suggested Edits:

Edit #1 | Clarity

Original: To compare to previous work, we focus on four data sets.

Suggested: To enable comparison with previous work, we focus on four datasets.

Reason: The suggested phrasing is more formal and clearly states the purpose. 'Dataset' is typically written as one word in this context.

Edit #2 | Clarity

Original: This data set was also used in ref 28 and adapted to a smaller subset called USPTO_LEF by Bradshaw et al. 25 to make it compatible with their algorithm.

Suggested: This dataset was also used in ref 28. Subsequently, Bradshaw et al. adapted it into a smaller subset, USPTO_LEF, to ensure compatibility with their algorithm.

Reason: Breaking the long sentence into two improves readability and clarifies the sequence of events.

Edit #3 | Grammar

Original: In contrast with the MIT and LEF data sets, USPTO_STEREO 28 underwent less ■ ltering, and the stereochemical information was kept.

Suggested: In contrast to the MIT and LEF datasets, USPTO_STEREO retained stereochemical information and underwent less filtering.

Reason: 'In contrast to' is the more common and preferred academic idiom. Rephrasing with 'retained' creates a more active and direct sentence structure.

Edit #4 | Tone

Original: To date, only seq-2-seq models were used to predict on USPTO_STEREO.

Suggested: To date, prediction on the USPTO_STEREO dataset has been performed exclusively using seq-2-seq models.

Reason: This revision is more formal and precise than the original phrasing.

Edit #5 | Clarity

Original: So far, in most of the work, the reagents have been separated from the reactants.

Suggested: In most prior work, reagents were separated from reactants.

Reason: This revision is more concise and direct, removing the slightly colloquial 'So far'.

Edit #6 | Tone

Original: In ref 26, it was shown that the model performs better when the reagents are tagged as such.

Suggested: For example, ref 26 demonstrated improved model performance when reagents were explicitly tagged.

Reason: Avoids the passive 'it was shown' and directly attributes the finding to the reference, creating a stronger, more academic sentence.

Edit #7 | Tone

Original: Unfortunately, the separation of reactants and reagents is not always obvious.

Suggested: However, the separation of reactants from reagents is not always unambiguous.

Reason: Replaces the slightly editorial 'Unfortunately' with the more neutral 'However' and uses the more precise term 'unambiguous' instead of 'obvious'.

Edit #8 | **Tone**

Original: We called this method of preprocessing mixed .

Suggested: We refer to this preprocessing method as 'mixed'.

Reason: 'We refer to' is more formal academic phrasing than 'We called'. Using quotes for the term 'mixed' clearly designates it as a specific label.

Edit #9 | **Clarity**

Original: The mixed preprocessing makes the reaction prediction task signi ■ cantly harder because the model has to determine the reaction center from a larger number of molecules.

Suggested: The 'mixed' preprocessing makes the reaction prediction task considerably more challenging, as the model must identify the reaction center from a larger set of molecules.

Reason: Replaces 'significantly'—which can imply statistical significance—with 'considerably' for emphasis. 'More challenging' is a strong alternative to 'harder'.

Edit #10 | **Clarity**

Original: All of the reactions used in this work were canonicalized using RDKit.

Suggested: All reactions in this work were canonicalized using RDKit.

Reason: The revision is more concise by removing the unnecessary phrase 'of the'.

Section 4: ■ MOLECULAR TRANSFORMER

Metric	Score	Visual
Clarity	7/10	■■■■■■■■■■■
Logic	8/10	■■■■■■■■■■■
Rigor	9/10	■■■■■■■■■■■

Summary:

The section provides a robust and detailed description of the Transformer architecture, the specific model configuration, and the training procedure. While scientifically rigorous and largely reproducible, it suffers from minor structural issues, occasional colloquialisms, and a few unsupported claims that should be tied directly to the student's results.

Critical Issues:

• Improve transitions between the general model description and the specific implementation details to create a smoother narrative flow. For example, add a sentence to bridge the explanation of positional encodings and the start of the implementation details (beam search).
• Integrate Table 1 into the text by explicitly stating that the described model was trained and evaluated on these datasets. The current placement is disjointed and lacks context.

• Substantiate the claim regarding the negative effect of label smoothing by referencing specific results (e.g., 'As shown in Table X...') or rephrase it as an empirical finding of this study, not a general fact.

Suggested Edits:

Edit #1 | Clarity

Original: The main architectural di ■ erence compared with seq-2-seq models previously used for the reaction prediction 27 , 28 is that the RNN component was completely removed, and it is fully based on the attention mechanism.

Suggested: Its main architectural difference from the seq-2-seq models previously used for reaction prediction [27, 28] is the complete removal of the Recurrent Neural Network (RNN) component in favor of a design based entirely on the attention mechanism.

Reason: Improves flow, uses more formal language ('in favor of'), and clarifies the acronym RNN. Using bracketed citations [27, 28] is a common and clean academic style.

Edit #2 | Tone

Original: It basically combines the information of the source sequence with the target sequence that has been produced so far.

Suggested: This layer integrates information from the source sequence with the target sequence generated thus far.

Reason: Removes the colloquialism 'basically' and uses the more precise term 'integrates'. 'Thus far' is more formal than 'so far'.

Edit #3 | Clarity

Original: The encoder computes interesting features from the input sequence, which are then queried by the decoder depending on its preceding outputs.

Suggested: The encoder generates a contextualized representation of the input sequence, which the decoder then queries based on its own preceding outputs.

Reason: Replaces the vague and unscientific term 'interesting features' with the more descriptive and standard 'contextualized representation'.

Edit #4 | Tone

Original: ...allows the encoder and decoder to peek at di ■ erent tokens simultaneously.

Suggested: ...allows the model to attend to multiple parts of the input sequence simultaneously.

Reason: Replaces the colloquialism 'peek at' with the correct technical term 'attend to' from the attention mechanism literature.

Edit #5 | Tone

Original: The top-k outputs are decoded via a beam search.

Suggested: We decode the top-k outputs using a beam search algorithm.

Reason: Switches from passive to active voice ('We decode...') for consistency with the surrounding sentences ('We set the beam size...').

Edit #6 | Clarity

Original: ...we decreased the number of trainable weights to 12 M by going from six layers of size 512 to four layers of size 256.

Suggested: ...we decreased the number of trainable weights to 12M by reducing the number of layers from six to four and the hidden dimension from 512 to 256.

Reason: More precise and formal phrasing than 'going from'. Explicitly names the parameters being changed ('number of layers', 'hidden dimension').

Edit #7 | Clarity

Original: As seen below, a nonzero label smoothing parameter encourages the model to be less con ■ dent and therefore negatively a ■ ects its ability to discriminate between correct and incorrect predictions.

Suggested: We observed that a non-zero label smoothing parameter ($\lambda > 0.0$) negatively affected the model's discrimination performance, as it encourages less confident predictions.

Reason: Frames the statement as an empirical finding of this specific study ('We observed...') rather than a universal fact. It removes the unsubstantiated 'As seen below' which points to missing evidence.

Edit #8 | Tone

Original: We, however, kept the original eight attention heads because this con ■ guration achieved superior validation performance.

Suggested: However, we retained the original eight attention heads, as this configuration achieved higher validation accuracy.

Reason: Improves sentence flow by moving 'However' to the beginning and replaces the vague 'superior validation performance' with a more specific and measurable metric like 'higher validation accuracy'.

Edit #9 | Clarity

Original: ...the batch size was set to ~ 4096 tokens, and the gradients were accumulated over four batches and normalized by the number of tokens.

Suggested: For training, we used the ADAM optimizer, a variable learning rate schedule with 8000 warm-up steps, and a batch size of approximately 4096 tokens. Gradients were accumulated over four steps and subsequently normalized by the total number of tokens.

Reason: Restructures the sentence for better readability, uses active voice and parallelism, and clarifies the gradient accumulation process.

Section 5: ■ RESULTS AND DISCUSSION

Metric	Score	Visual
Clarity	9/10	■■■■■■■■■■
Logic	10/10	■■■■■■■■■■
Rigor	9/10	■■■■■■■■■■

Summary:

This section is exceptionally strong, presenting a comprehensive and logically structured validation of the Molecular Transformer model. The argument flows seamlessly from quantitative ablation studies and state-of-the-art comparisons to insightful qualitative analyses and an investigation of model properties like uncertainty estimation. The scientific rigor is high, demonstrated by fair comparisons, critical analysis of failure cases, and well-designed experiments. The primary areas for improvement are minor: refining the academic tone by removing colloquialisms and ensuring the term 'significant' is reserved for statistically validated claims.

Critical Issues:

• The term 'significant' is used repeatedly to describe performance increases without providing statistical validation (e.g., p-values). This term should be reserved for statistically significant results or replaced with more descriptive, qualitative language (e.g., 'notable', 'substantial', 'measurable').
• The section contains several instances of informal or colloquial language ('looming question', 'ominous label', 'alchemy') that slightly detract from the authoritative academic tone. While sometimes effective for illustration, they should be used sparingly or replaced with more formal terminology.
• The structure is excellent, but some paragraphs combine multiple distinct points. For instance, the first paragraph discusses data augmentation, weight averaging, and ensembling. While related, breaking these into more focused paragraphs or using clearer topic sentences could improve readability further.

Suggested Edits:

Edit #1 | **Tone**

Original: SMILES data augmentation 46 leads to a signi ■ cant increase in accuracy.

Suggested: SMILES data augmentation 46 leads to a notable increase in accuracy.

Reason: The term 'significant' in academic writing implies statistical significance. Since no statistical test is mentioned, 'notable' or 'measurable' is a more precise and defensible descriptor for the observed increase from 88.8% to 89.6%.

Edit #2 | **Clarity**

Original: Crucially, although separating reactant and reagent yields, the best model (perhaps unsurprisingly because this separation implies knowledge of the product already), the Molecular Transformer, still outperforms the literature when reactant and reagents are mixed.

Suggested: Crucially, the Molecular Transformer outperforms literature methods even when reactants and reagents are mixed. This is a more challenging task, as separating them provides an implicit hint about the reaction, which may explain the superior performance of models using separated inputs.

Reason: The original sentence is long and convoluted, with a parenthetical that disrupts the flow. The revision breaks the idea into two clearer, more direct sentences, improving readability while retaining the core meaning.

Edit #3 | **Tone**

Original: A looming question is how the Molecular Transformer performs by reaction type.

Suggested: A key question is how the Molecular Transformer's performance varies by reaction type.

Reason: The phrase 'A looming question' is overly dramatic and colloquial for a scientific paper. 'A key question' or 'An important consideration' is more formal and standard academic phrasing.

Edit #4 | **Tone**

Original: ...and the ominous label of “ unclassi ■ ed ” (where many mistranscribed reactions will end up).

Suggested: ...and reactions labeled as “unclassified,” a category that likely contains many mistranscribed reactions.

Reason: The word 'ominous' is subjective and informal. The revision is more neutral and objective, simply stating the likely composition of the 'unclassified' category.

Edit #5 | Clarity

Original: This is because the multihead attention layer in the Molecular Transformer can process long-range interactions between tokens, whereas RNN models impose the inductive bias that tokens far in sequence space are less related.

Suggested: We attribute this to the multihead attention layer in the Molecular Transformer, which can process long-range interactions between tokens. In contrast, RNN models impose an inductive bias that assumes tokens distant in the sequence are less related.

Reason: Starting with 'We attribute this to...' is a stronger, more assertive academic stance than 'This is because...'. Splitting the contrast into a separate sentence improves readability and emphasizes the difference between the model architectures.

Edit #6 | Tone

Original: Surprisingly, Molecular Transformer achieves a top-1 accuracy of 83% and top-2 accuracy of 91% on the same data set...

Suggested: Notably, the Molecular Transformer achieves a top-1 accuracy of 83% and a top-2 accuracy of 91% on the same dataset...

Reason: 'Surprisingly' is a subjective word that expresses the author's emotion. 'Notably' or simply stating the fact without an adverb is more objective and allows the result to speak for itself.

Edit #7 | Grammar

Original: However, Figure 7 shows that this small increase in accuracy comes at the cost of no longer being to able to discriminate between a good and a bad prediction.

Suggested: However, Figure 7 shows that this small increase in accuracy comes at the cost of being unable to discriminate between a good and a bad prediction.

Reason: The phrase 'to able to' is grammatically incorrect. The correct phrasing is 'able to' or, in this context, the more concise 'being unable to'.

Edit #8 | Clarity

Original: As such, an interesting question is whether the model has learned to avoid alchemy.

Suggested: This raises the question of whether the model has learned fundamental chemical constraints, such as the conservation of atoms (i.e., avoiding 'alchemy').

Reason: While 'alchemy' is a clever and illustrative term, explicitly defining it in the context of atom conservation makes the scientific point clearer and more rigorous for all readers, while retaining the memorable concept.

Section 6: ■ CONCLUSIONS

Metric	Score	Visual
Clarity	6/10	■■■■■■■■■■
Logic	8/10	■■■■■■■■■■
Rigor	5/10	■■■■■■■■■■

Summary:

This section effectively summarizes the key achievements of the Molecular Transformer model, showcasing its high accuracy and real-world impact. However, its academic rigor is undermined by unsubstantiated superlative claims, vague terminology for experimental setups, and a lack of specific citations for benchmark datasets and competing methods.

Critical Issues:

• Unsubstantiated Superlative Claims: The text asserts the model 'outperforms all known algorithms' without providing citations or a comparative table to support this strong statement. Such claims require robust evidence.
• Vague Experimental Details: Key elements like the 'common benchmark data set' are not named, and the description of results on the 'USPTO_STEREO data set' is ambiguous, hindering reproducibility and verification.
• Subjective and Informal Language: Phrases like 'unrealistically generous' and 'we point out' detract from the objective, academic tone required in a thesis. Claims should be framed neutrally and factually.

Suggested Edits:

Edit #1 | Clarity

Original: We show that a multihead attention Transformer network, the Molecular Transformer, outperforms all known algorithms in the reaction prediction literature, achieving 90.4% top-1 accuracy (93.7% top-2 accuracy) on a common benchmark data set.

Suggested: This work demonstrates that a multihead attention Transformer network, the 'Molecular Transformer', outperforms state-of-the-art algorithms for reaction prediction, achieving 90.4% top-1 accuracy on the USPTO-50k benchmark dataset [citation needed].

Reason: Replaces the informal 'We show' with 'This work demonstrates'. 'All known algorithms' is an unverifiable superlative; 'state-of-the-art algorithms' is more conventional and defensible. The 'common benchmark data set' must be explicitly named for scientific rigor and reproducibility, and a citation should be provided.

Edit #2 | Clarity

Original: The model requires no handcrafted rules and accurately predicts subtle chemical transformations.

Suggested: The model, which requires no handcrafted rules, accurately predicts complex chemical transformations not easily captured by rule-based systems.

Reason: The term 'subtle' is subjective. 'Complex' is a more standard descriptor. The revision adds context by contrasting the model's performance with the rule-based systems it aims to surpass.

Edit #3 | **Clarity**

Original: Moreover, the Molecular Transformer can also accurately estimate its own uncertainty, with an uncertainty score that is 89% accurate in terms of classifying whether a prediction is correct.

Suggested: Furthermore, the Molecular Transformer accurately estimates its own uncertainty, achieving 89% classification accuracy when predicting whether its own output is correct.

Reason: Replaces the slightly redundant 'can also' and clarifies the clunky phrase 'in terms of classifying whether a prediction is correct' with a more direct and concise statement.

Edit #4 | **Tone**

Original: We point out that previous work has considered an unrealistically generous setting of separated reactants and reagents.

Suggested: This analysis addresses a limitation in prior work, which often relied on a simplified setting where reactants and reagents are pre-separated as distinct inputs.

Reason: Replaces the informal 'We point out' and the subjective judgment 'unrealistically generous' with more neutral, academic language that frames the contribution constructively.

Edit #5 | **Citation**

Original: We demonstrate an accuracy of 88.6% when no distinction is drawn between reactants and reagents in the inputs, a score that outperforms previous work as well.

Suggested: In this more challenging setting, the model achieves an 88.6% top-1 accuracy, a result that exceeds the performance of previously reported methods [citation needed].

Reason: The claim of 'outperforming previous work' is a critical assertion that must be substantiated with a specific citation to the work being compared against.

Edit #6 | **Clarity**

Original: For the more noisy USPTO_STEREO data set, our top-1 accuracies are 78.1 (separated) and 76.2%, respectively.

Suggested: On the more complex USPTO_STEREO dataset, the model achieves top-1 accuracies of 78.1% (with separated inputs) and 76.2% (with mixed inputs).

Reason: Replaces the informal term 'noisy' with 'complex' or 'challenging'. The revision explicitly defines the two conditions being tested, resolving the ambiguity of '(separated)' and the confusing use of 'respectively'.

Edit #7 | **Tone**

Original: The Molecular Transformer has been freely available since August 2018 through a graphical user interface on the IBM RXN for Chemistry platform 34 and has so far been used by several thousand organic chemists worldwide to perform more than 40 000 chemical reaction predictions.

Suggested: The Molecular Transformer has been publicly available via a graphical user interface on the IBM RXN for Chemistry platform [34] since August 2018, where it has been used to perform over 40,000 reaction predictions for thousands of chemists worldwide.

Reason: Removes the informal and temporal phrase 'so far'. Standardizes the citation format to brackets and rephrases slightly for a more formal flow.

Section 7: ■ AUTHOR INFORMATION

Metric	Score	Visual
Clarity	9/10	■■■■■■■■■■
Logic	8/10	■■■■■■■■■■
Rigor	9/10	■■■■■■■■■■

Summary:

This section appears to be an excerpt from the end of a typeset, published article, containing author metadata, a conflict of interest statement, and figure captions. The content is not a narrative section of a thesis (e.g., Introduction, Discussion). The figure captions are a key strength, demonstrating high scientific rigor and clarity by providing specific details about the models, datasets, and key statistical findings, making the figures highly self-contained. The primary weakness is the presence of typesetting artifacts (ligatures like '■', '■'), which should be resolved in a manuscript draft. The logical structure is appropriate for its context within a journal publication.

Critical Issues:

• No critical structural or scientific flaws were identified in the provided text.
• The text is a fragment from a final publication, not a manuscript draft, which limits the scope of a traditional thesis review. The feedback focuses on the content of the figure captions and philological corrections.
• The text contains numerous typesetting ligatures (e.g., '■', '■', '■') that must be corrected to standard characters ('fi', 'ff', 'ffi') for a clean manuscript.

Suggested Edits:

Edit #1 | Clarity

Original: The authors declare no competing ■ nancial interest.

Suggested: The authors declare no competing financial interest.

Reason: Corrects the typesetting ligature '■' to 'fi'. This ensures text consistency and avoids rendering issues in plain-text editors or different software.

Edit #2 | Clarity

Original: Receiver operating characteristic curve for di ■ erent label smoothing values for a model trained on the mixed USPTO_MIT data set when evaluated on the validation set.

Suggested: Receiver operating characteristic (ROC) curves for different label smoothing values (e.g., 0.0, 0.1, 0.2) for a model trained on the mixed USPTO_MIT dataset and evaluated on the validation set.

Reason: Introduces the standard acronym (ROC) for future reference, corrects the '■' ligature, and suggests adding specific examples of the 'different values' to enhance precision, assuming they are shown in the figure.

Edit #3 | Clarity

Original: The Pearson product moment correlation coe ■ cient between the length and the con ■ dence score is 0.06.

Suggested: The Pearson product-moment correlation coefficient between the sequence length and the confidence score is 0.06.

Reason: Corrects the ligatures '■' and '■'. Also clarifies 'length' to 'sequence length' for explicitness and standardizes 'product moment' to the more common hyphenated form 'product-moment'.

Section 8: ■ ACKNOWLEDGMENTS

Metric	Score	Visual
Clarity	9/10	■■■■■■■■■■■
Logic	10/10	■■■■■■■■■■■
Rigor	9/10	■■■■■■■■■■■

Summary:

This section is a concise and well-structured acknowledgement of funding and intellectual contributions, adhering to academic conventions. Minor revisions can enhance consistency in personal voice and ensure typographical accuracy.

Critical Issues:

• Inconsistent authorial voice, shifting from third-person initials to the first-person plural ('We').
• Presence of a typographic ligature ('■') that may render incorrectly and should be standardized.

Suggested Edits:

Edit #1 | **Tone**

Original: P.S. and A.A.L. acknowledge the Winton Programme for the Physics of Sustainability for funding. We thank G. Landrum, R. Sayle, G. Godin, and R. Gri ■ ths for useful feedback and discussions.

Suggested: We acknowledge funding from the Winton Programme for the Physics of Sustainability. We also thank G. Landrum, R. Sayle, G. Godin, and R. Griffiths for their useful feedback and discussions.

Reason: This revision establishes a consistent first-person plural voice ('We'), which is more conventional for a thesis than using author initials. It resolves the inconsistency between the first and second sentences, improves flow by adding 'also', and corrects the typographic ligature in 'Griffiths' to 'ffi' to ensure proper rendering. The phrasing 'funding from' is also slightly more direct.

Section 9: ■ REFERENCES

Metric	Score	Visual
Clarity	7/10	■■■■■■■■■■■
Logic	9/10	■■■■■■■■■■■
Rigor	6/10	■■■■■■■■■■■

Summary:

The reference list is comprehensive and topically coherent, drawing from high-impact journals and conferences in both chemistry and machine learning. This indicates a strong foundation for the research. The primary weakness is the inconsistent and often incorrect formatting of the citations, which undermines the section's scholarly polish and must be systematically corrected to adhere to a single, recognized academic style (e.g., ACS style).

Critical Issues:

• Inconsistent Citation Formatting: The list mixes different formatting rules for journals, conference proceedings, patents, and web resources. A single, consistent style (e.g., ACS) must be applied throughout.
• Missing Essential Information: Many journal entries lack complete page ranges or Digital Object Identifiers (DOIs), which are standard and crucial for traceability. This must be rectified for every entry.
• Capitalization and Punctuation Errors: There are numerous errors in the capitalization of article and journal titles, as well as incorrect or missing punctuation (commas, periods) that deviate from standard academic citation practices.

Suggested Edits:

Edit #1 | Clarity

Original: (7) Popova, M.; Isayev, O.; Tropsha, A. Deep Reinforcement Learning for De Novo Drug Design. Science advances 2018 , 4 , No. eaap7885.

Suggested: (7) Popova, M.; Isayev, O.; Tropsha, A. Deep Reinforcement Learning for De Novo Drug Design. *Sci. Adv.* **2018**, *4*, eaap7885.

Reason: Corrects the journal title to its standard abbreviation and capitalization (*Sci. Adv.*). The year should be bolded, and the volume number italicized, per ACS style. The 'No.' prefix is redundant for an article number.

Edit #2 | Clarity

Original: (16) Segler, M. H.; Waller, M. P. Neural-Symbolic Machine Learning for Retrosynthesis and Reaction Prediction. Chem. - Eur. J. 2017 , 23 , 5966 – 5971.

Suggested: (16) Segler, M. H.; Waller, M. P. Neural-Symbolic Machine Learning for Retrosynthesis and Reaction Prediction. *Chem. Eur. J.* **2017**, *23*, 5966–5971.

Reason: Standardizes formatting: the journal title should be italicized without the extra spaces and hyphen, the year should be bold, and the volume number should be italicized. The en-dash (–) is preferred for page ranges over a spaced hyphen (-).

Edit #3 | Clarity

Original: (27) Nam, J.; Kim, J. Linking the Neural Machine Translation and the Prediction of Organic Chemistry Reactions. 2016, arXiv:1612.09529. arXiv.org e-Print archive. https://arxiv.org/abs/1612.09529 (accessed July 29, 2019).

Suggested: (27) Nam, J.; Kim, J. Linking the Neural Machine Translation and the Prediction of Organic Chemistry Reactions. arXiv:1612.09529 [cs.LG], 2016. (accessed Jul 29, 2019).

Reason: Streamlines the citation for a pre-print server. The redundant URL and 'e-Print archive' text are removed in favor of the standard arXiv identifier format. The date format is also standardized.

Edit #4 | Citation

Original: (34) IBM RXN for Chemistry . https://rxn.res.ibm.com (accessed July 29, 2019).

Suggested: (34) IBM. *RXN for Chemistry*. https://rxn.res.ibm.com/ (accessed Jul 29, 2019).

Reason: Attributes the work to the parent organization (IBM) as the author and italicizes the title of the web resource, following common citation practices for software and websites. A period should follow the title.

Edit #5 | Citation

Original: (57) Papineni, K.; Roukos, S.; Ward, T.; Zhu, W.-J. BLEU: A Method for Automatic Evaluation of Machine Translation. Proceedings of the 40th Annual Meeting on Association for Computational Linguistics 2002 , 311 – 318.

Suggested: (57) Papineni, K.; Roukos, S.; Ward, T.; Zhu, W.-J. BLEU: A Method for Automatic Evaluation of Machine Translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*; Association for Computational Linguistics: Stroudsburg, PA, USA, 2002; pp 311–318.

Reason: Adopts a standard, more complete format for conference proceedings. This includes specifying the publisher and location, which improves traceability and scholarly rigor. The 'pp' prefix for page numbers is standard in this context.

Edit #6 | Citation

Original: (59) Segler, M. H.; Preuss, M.; Waller, M. P. Planning Chemical Syntheses with Deep Neural Networks and Symbolic AI. Nature 2018 , 555 , 604.

Suggested: (59) Segler, M. H.; Preuss, M.; Waller, M. P. Planning Chemical Syntheses with Deep Neural Networks and Symbolic AI. *Nature* **2018**, *555*, 604–609.

Reason: Provides the complete page range (604–609), which is essential, instead of only the starting page. Also applies standard ACS formatting (italicized journal and volume, bolded year) for consistency.